Sub-pixel spot localization under sensor noise

Abstract

This report evaluates sub-pixel laser-spot localization for a camera-guided pointing and tracking system operating near a 1 kHz frame rate. A configurable simulator combines a pixel-integrated circular Gaussian point-spread function with photon shot noise, read noise, dark current, hot pixels, background gradients, pixel-response non-uniformity, quantization, and saturation. A windowed, background-subtracted intensity centroid is compared with a bounded pixel-integrated two-dimensional Gaussian fit using Monte Carlo experiments and a clean-model Fisher-information reference.

The central result is that no estimator dominates across all operating conditions. In the clean low-SNR case, the Gaussian fit had higher lock loss at SNR 1 than the centroid (26.0% versus 7.2%), whereas under the stressed sensor model the ranking reversed (90.1% versus 96.5%). When the spot was observable at SNR 10, the Gaussian fit achieved lower stressed-profile radial RMSE than the centroid (0.081 versus 0.161 pixels), but its Python implementation did not meet the 1 ms estimator budget, while the centroid remained below that budget on both the Mac reference system and Jetson Orin. The dynamic experiment recovered the injected 73 Hz disturbance from a 5,000-frame trajectory, and a controlled outdoor-stress study examined scintillation, illumination, defocus, and saturation as distinct failure mechanisms. An optional live GUI that uses the same simulator and estimators is developed to demonstrate these regime changes interactively.

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1. Introduction

The application considered in this exercise is a pair of camera-guided, gimbal-mounted laser platforms. A camera observes the laser spot produced by the opposing platform, and the measured spot position is supplied to a stabilization

loop. Platform vibration, optical variation, illumination, and sensor noise make the measurement problem non-trivial. The objective of this exercise is to evaluate the accuracy and computational cost of sub-pixel spot localization under controlled simulated conditions.

The assignment requires four core elements: a configurable sensor simulator, at least two localization estimators, Monte Carlo characterization as a function of SNR, and per-frame runtime analysis. It additionally requires a moving-spot experiment containing slow drift, random jitter, and one periodic disturbance. The present report follows that structure. The complete methodology is summarized in `docs/methodology_flowchart.md`.

2. Coordinate system and sensor model

2.1 Coordinate and unit conventions

Images are stored as `image[y, x]`. Cartesian spot positions are represented as `(x, y)`. Pixel centers have integer coordinates, and pixel boundaries are located half a pixel from the corresponding center. Charge quantities are represented in electrons until conversion to analogue-to-digital units (DN). The nominal sensor contains 32×32 pixels, has a 1 ms exposure, and operates at a nominal frame rate of 1 kHz.

The spot width is specified using the assignment's approximately 7-pixel diameter at the $1/e^2$ intensity level. Let w denote the corresponding $1/e^2$ radius, so the assignment diameter gives $w = 3.5$ pixels. For

$$I(r) = I_0 \exp\left(-\frac{2r^2}{w^2}\right),$$

the corresponding Gaussian standard deviation is $\sigma = w/2$. Thus, a 7-pixel diameter gives a nominal $\sigma = 1.75$ pixels.

2.2 Pixel-integrated point-spread function

The expected spot image is generated by integrating a normalized circular Gaussian over the finite area of each pixel. The separability of the Gaussian allows the two-dimensional integral to be evaluated as the outer product of two one-dimensional normal-CDF differences. This preserves sub-pixel phase effects and avoids the bias that would result from evaluating the Gaussian at pixel centers only.

The physical brightness parameter is the total expected spot charge, N_{total}, in electrons. The simulator also accepts a target SNR, which is converted to N_{total} using a reference spot centered on a pixel. Because the actual peak pixel depends on sub-pixel position and spot width, each rendered frame records both the reference SNR and an effective SNR.

2.3 Sensor and noise model

The nominal stressed-sensor configuration uses a 12-bit ADC, 1 electron/DN conversion gain, a 4095-electron full-well limit, and 3 electrons RMS read noise. The background is a planar field with a 20-electron offset, an x gradient of 0.25 electrons per pixel, and a y gradient of -0.15 electrons per pixel. The default fixed-pattern model uses 1% PRNU, a 50-electron/s baseline dark-current rate, and four hot pixels with a 5,000-electron/s dark-current rate.

The sensor chain is applied in the following order:

1. generate expected spot, background, and dark-current charge;
2. apply a persistent multiplicative PRNU gain map;
3. sample the total expected charge using independent Poisson draws;
4. add independent Gaussian read noise in electrons;
5. combine the full-well and ADC limits into an equivalent charge limit and clip the charge in electrons;
6. convert the clipped charge from electrons to DN;
7. apply round-to-nearest quantization and the final DN-range clamp when enabled.

The conversion gain is explicit even though the nominal configuration uses 1 electron/DN, so the numerical DN and electron values coincide in the main run. Full-well clipping and the ADC limit are both converted to an equivalent charge limit before conversion. A unit test repeats this check with a 2-electron/DN gain and separate full-well and ADC limits; this prevents the nominal 1-electron/DN choice from hiding a unit-ordering error.

The simulator's clipping mask records both high-side saturation and low-side negative excursions from additive read noise before ADC conversion. In the high-SNR dynamic result, the reported saturation behavior is dominated by high-side spot clipping; low-side clipping is a separate readout artifact.

Photon noise, read noise, dark current, background, PRNU, and hot pixels can be enabled independently through `NoiseConfig`. Quantization and saturation are controlled through `SensorConfig`. Fixed-pattern maps persist for the lifetime of a simulator and therefore persist across a dynamic sequence.

3. Localization estimators

Both estimators use the full image only to locate the brightest finite pixel. They then operate on a local, odd-sized ROI and never use the simulator's true position. The default centroid ROI is 13×13 pixels and the default Gaussian fit ROI is 15×15 pixels.

3.1 Background-subtracted intensity centroid

The centroid estimator fits either a constant background or a plane to the border of the local ROI:

$$B(x, y) = b_0 + b_x(x - c_x) + b_y(y - c_y).$$

The fitted background is subtracted from the ROI, negative residuals are clipped to zero, and the remaining intensities are used as weights:

$$\hat{x} = \frac{\sum_i w_i x_i}{\sum_i w_i}, \qquad \hat{y} = \frac{\sum_i w_i y_i}{\sum_i w_i}.$$

The estimator returns the position, ROI bounds, peak value, saturation flag, positive signal, and fitted background coefficients. It returns an explicit failure result when the image contains no usable signal rather than returning an arbitrary bounded coordinate.

The centroid is computationally simple and interpretable. It is nevertheless sensitive to background-model error, hot pixels, clipping, ROI truncation, and the removal of faint PSF tails during background subtraction.

3.2 Pixel-integrated Gaussian fit

The second estimator fits the parameter vector

$$\theta = (x_0, y_0, A, \sigma, b_0, b_x, b_y),$$

where A is total spot flux in DN and the background is a local plane. The spot contribution is integrated over each pixel using normal-CDF differences, matching the simulator's forward model. Position, flux, and width are bounded by configurable limits. Background coefficients remain unbounded.

The initial position is obtained from the centroid when available and from the peak pixel otherwise. The fit uses bounded SciPy `least_squares` with an analytical Jacobian. Two fixed iteratively reweighted steps approximate Poisson-plus-read-noise weighting using

$$V_i \approx \max\left(\frac{\mu_i}{g}, V_{\text{floor}}\right) + \sigma_{r,DN}^2,$$

where g is the electrons-per-DN conversion gain and V_{floor} is the configurable DN^2 variance floor. The floor prevents tiny or negative fitted tail-background values from receiving numerically dominant weights. Pixels at the ADC limit are excluded when saturation is enabled. The implementation records optimizer cost, function evaluations, optimality, fitted width and flux, background coefficients, valid-pixel count, and saturation count.

The fit retains an acquisition mode based on full-frame peak detection and a tracking mode that can reuse the previous estimate to place the ROI. The tracking mode can also skip the redundant centroid initialization. ROI

size, IRLS pass count, optimizer iteration limit, and these tracking choices are configuration parameters rather than separate estimator implementations.

This method uses more information than the centroid and can estimate the spot shape and background jointly. It is also more computationally expensive and can fail under low SNR, poor initialization, saturation, or model mismatch. It is therefore described as a noise-weighted nonlinear fit, or quasi-MLE, rather than an exact likelihood for clipped and quantized data.

4. Verification of the current implementation

The current test suite contains 44 tests. The tests are deterministic and use statistical tolerances where random sampling is involved. They cover:

- pixel-integrated PSF flux conservation and symmetry;
- sub-pixel position response and coordinate conventions;
- scenario and configuration validation;
- independent reproducibility of named random-number-generator streams;
- persistent PRNU, dark-current, and hot-pixel maps;
- Poisson and read-noise sample statistics;
- SNR-to-flux monotonicity and reference-SNR inversion;
- ADC quantization, clipping, and saturation invariants;
- independent noise-component ablations;
- background-gradient orientation and exposure scaling;
- composed Poisson/read-noise behavior and edge behavior;
- clean centroid and Gaussian-fit recovery;
- analytical Gaussian Jacobian agreement with finite differences;
- explicit failure on unusable frames;
- saturation handling in the Gaussian fit;
- reproducibility and bounds of the trajectory generator;
- headless rendering of one live-GUI update;
- characterization metrics, lock-loss classification, and the theoretical precision reference;
- runtime benchmark output and latency-statistic formatting;
- OU trajectory bounds/reproducibility and dynamic disturbance recovery;
- reproducibility and diagnostic ordering of the controlled real-world condition analysis.

The clean Gaussian-fit test recovers the injected sub-pixel position to numerical precision, while the centroid test uses a looser tolerance appropriate to finite ROI background subtraction. These tests verify the implementation but are not substitutes for the required Monte Carlo characterization.

5. Static accuracy characterization

The static characterization runner uses Monte Carlo trials over the configured SNR sweep. The clean reference condition holds the PSF width and background fixed and disables clipping and quantization. The stressed condition enables the configured sensor effects and sigma variation. Spot positions are sampled within the safe frame margin, and the same positions, fixed-pattern maps, and rendered frames are used for both estimators at each SNR.

For each estimator and SNR, the following quantities are reported for both axes and radially where appropriate:

- signed bias;
- standard deviation;
- RMSE;
- median and 95th-percentile absolute error;
- invalid-result, failure, and outlier rates;
- x/y error covariance and correlation.

In addition to unconditional statistics, a lock-loss threshold of 1 pixel is used for simulation analysis. A formal estimator failure is an invalid result or optimizer failure; a lock loss is either such a failure or a finite result whose radial error exceeds the threshold. Conditional bias and precision are reported only for locked trials. This truth-based lock classification is appropriate for evaluating the simulator, but a deployed system would need operational quality flags rather than access to the true position.

The plots show uncertainty bars wherever a Monte Carlo uncertainty estimate is well-defined. Bias bars are standard errors of the mean over valid estimates; the per-axis standard-deviation bars use the usual approximation $s/\sqrt{2(n-1)}$. Radial RMSE bars use a delta-method standard error based on the sample variance of squared radial errors. Invalid-result and lock-loss bars are binomial standard errors computed over all requested trials. Thus the bars describe finite-trial uncertainty in the reported summary, rather than a confidence interval for an individual frame or a guarantee of estimator performance on hardware. The conditional statistics exclude lock-loss trials, while the unconditional RMSE and failure panels retain them in their stated denominators.

The required error-versus-SNR plots are generated by the offline characterization pipeline. The live GUI's position-error time history is a diagnostic display and does not replace these plots. A stressed-sensor ROI sensitivity study was run at SNR 3, 10, and 30 using 200 trials per setting. The 13×13 centroid ROI remains the selected compromise: a 15×15 ROI reduces some truncation error but increases low-SNR lock loss. The 15×15 Gaussian ROI is retained because it gives slightly lower high-SNR error and similar or lower lock loss than the smaller choices.

The final ideal and stressed sweeps used 1,000 trials at each of the seven configured SNR values. They produced machine-readable CSV and JSON output together with four-panel figures. The ideal-profile repeat produced byte-identical output, confirming deterministic reproduction. At SNR 1 in the stressed profile, lock loss was 96.5% for the centroid and 90.1% for the Gaussian fit. At SNR 3, the corresponding rates were 21.8% and 7.3%; both estimators had zero lock loss at SNR 10 and above until centroid saturation caused lock loss at SNR 300 and 1000. The Gaussian fit remained locked in those high-flux cases because saturated pixels were masked.

Profile	SNR	Centroid radial RMSE [px]	Gaussian radial RMSE [px]	Centroid lock loss	Gaussian lock loss
Ideal	1	0.635	0.900	7.2%	26.0%
Ideal	10	0.070	0.059	0.0%	0.0%
Ideal	1000	0.020	0.0006	0.0%	0.0%
Stressed	1	12.757	14.438	96.5%	90.1%
Stressed	3	4.239	4.811	21.8%	7.3%
Stressed	10	0.161	0.081	0.0%	0.0%
Stressed	1000	4.154	0.009	100.0%	0.0%

Figure 1. Bias, precision, radial RMSE, and lock-loss behavior across the SNR sweep for the stressed sensor profile. Error bars represent the finite- trial uncertainty described above.

5.1 Controlled explanation of the ranking reversal

The clean and stressed rankings are not expected to be identical. A small controlled ablation was run at SNR 1 and 3 with 250 paired trials per condition. At SNR 1, the clean model produced lock-loss rates of 10.8% for the centroid and 28.8% for the Gaussian fit, whereas the full stressed model produced 95.6% and 89.2%, respectively. At SNR 3, the stressed rates were 25.2% and 9.2%. These values reproduce the direction of the reversal in the 1,000-trial sweep rather than treating it as a plotting artefact.

The isolated effects identify the mechanism. At SNR 3, adding the planar background increased lock loss to 21.2% for the centroid and 9.2% for the Gaussian fit; adding PRNU and hot pixels gave 5.6% and 1.2%. Quantization and clipping alone were indistinguishable from the clean low-SNR result because the spot did not reach the ADC limit in that regime. The Gaussian fit is not universally more robust: at SNR 1 its nonlinear initialization and extra nuisance parameters make it less reliable in the clean photon-limited case, while under moderate contamination its fitted background plane and use of the full spot shape reduce the chance of accepting a noise/outlier-driven position. At severe low SNR both methods lose lock; the stressed Gaussian advantage is therefore a relative reduction in lock loss, not successful localization of most SNR-1 frames.

The ablation also records Gaussian fit cost and estimator disagreement for each condition. Its machine-readable outputs are in `results/latest/results/estimator_ablation/`, with the summary figure below.

Figure 2. Controlled lock-loss and RMSE ablation at SNR 1 and 3. Colors identify sensor conditions; circles and squares identify centroid and Gaussian estimates.

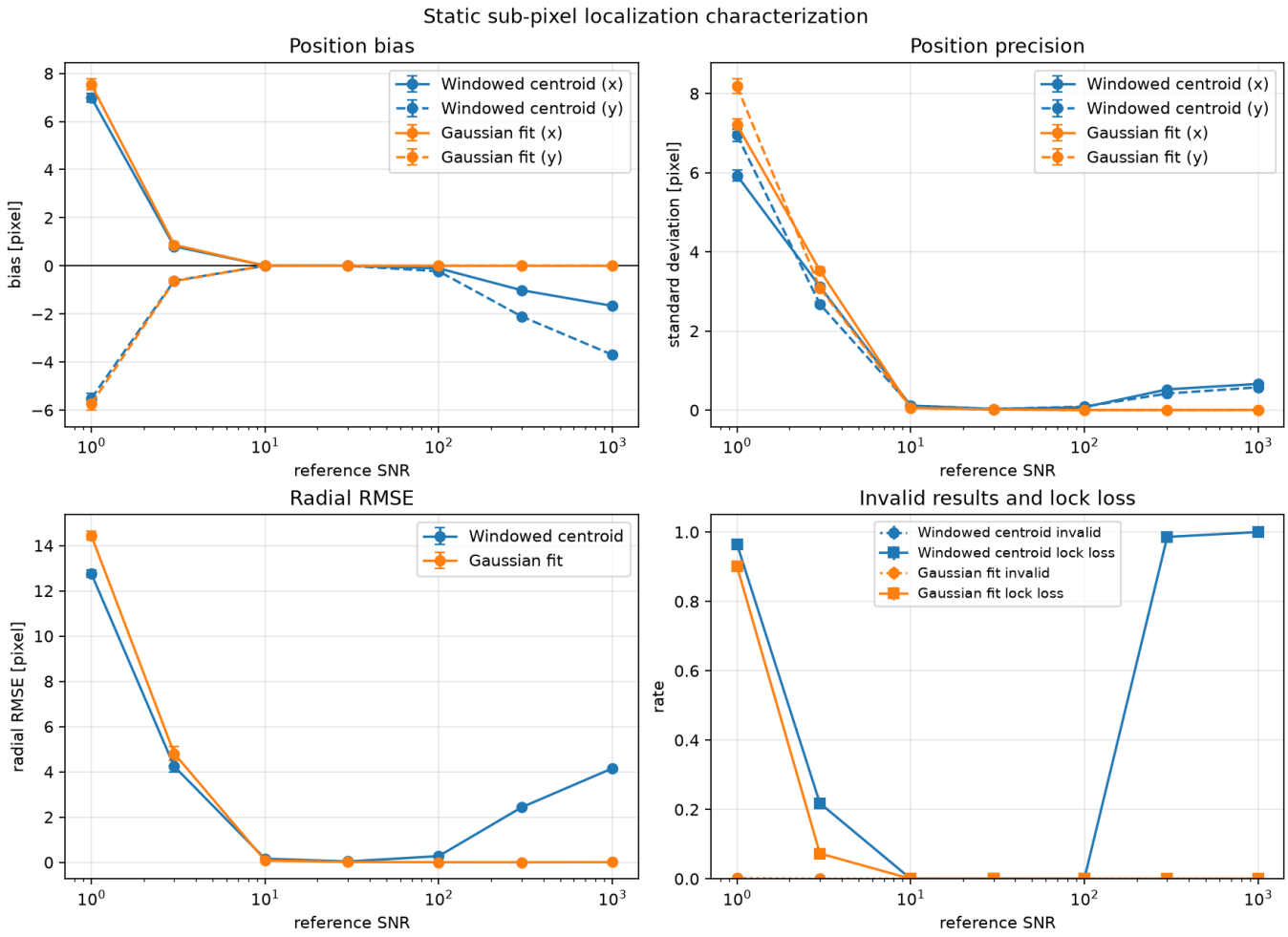


Figure 1: Stressed-sensor localization characterization

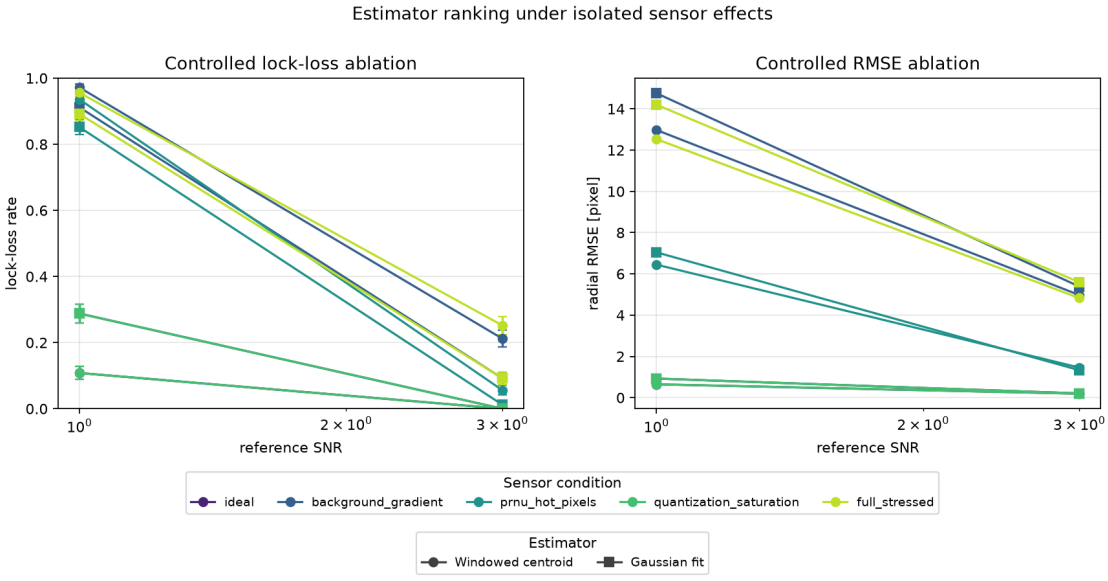


Figure 2: Controlled estimator ablation

6. Theoretical precision reference

A Fisher-information reference has been computed from the pixel-integrated forward model. The nuisance-parameter vector includes position, flux, width, and planar-background terms. For each pixel, the expected mean is

$$\mu_{ij} = A p_{ij}(x_0, y_0, \sigma) + b_0 + b_x(x_i - c_x) + b_y(y_j - c_y),$$

where p_{ij} is the product of the two normal-CDF differences over the finite pixel aperture. The implementation uses the diagonal Poisson-plus-read-noise variance approximation

$$V_{ij} \approx \frac{\max(\mu_{ij}, 0)}{g} + \sigma_{r,\mathrm{DN}}^2, \qquad \mathcal{J} = J^{\mathsf{T}} \mathrm{diag}(V_{ij}^{-1}) J,$$

where g is the electrons-per-DN conversion gain. The position covariance is obtained from the pseudoinverse of $\mathcal{J}$, including the nuisance parameters. The estimator's separate V_{floor} is not applied to this clean reference because the expected clean pixel means are non-negative. This is a clean-model Fisher reference (often called a quasi-CRLB here), not the exact Fisher information of the full Poisson–Gaussian convolution with parameter-dependent variance. The reference uses a configurable read-noise term, set to zero in the ideal profile, a 15×15 evaluation ROI, and an 8×8 grid of sub-pixel phases. The x/y variances are averaged over phase before taking the square root.

For the ideal profile, the resulting radial reference standard deviations are 0.564 pixels at SNR 1, 0.0564 pixels at SNR 10, and 0.000564 pixels at SNR 1000. The reference is overlaid on the position-precision panel of the ideal characterization figure. It is a precision reference for the stated model, not a universal bound for the stressed results.

This reference is a precision floor for the stated idealized approximation. ADC quantization, saturation, hot-pixel contamination, and unmodeled optical or sensor effects will be treated as model mismatch rather than implicitly included in the clean bound.

Figure 3. Ideal-model estimator comparison with the Fisher-information precision reference.

7. Real-time performance

Runtime was measured on 200 already-generated frames at target SNR 10, with 20 warm-up calls excluded from the statistics. This separates estimator cost from simulation cost. The nominal frame interval is 1 ms. The benchmark records cold-start centroid timing, cold-start Gaussian-fit timing, and Gaussian timing initialized from the previous estimate. It also records failure rate, 1 ms deadline misses, p95/p99 latency, and optimizer function evaluations.

The measurements were obtained with Python 3.13.1, NumPy 2.5.3, SciPy 1.18.1 on the recorded macOS arm64 development system.

Profile	Estimator	Initialization	Median [ms]	p95 [ms]	p99 [ms]	Worst [ms]	>1 ms
Ideal	Centroid	cold	0.054	0.064	0.067	0.069	0%
Ideal	Gaussian	cold	1.733	1.900	2.072	2.095	100%
Ideal	Gaussian	warm	1.677	1.885	2.054	2.249	100%
Stressed	Centroid	cold	0.055	0.061	0.065	0.071	0%
Stressed	Gaussian	cold	1.887	2.126	2.249	2.399	100%
Stressed	Gaussian	warm	1.819	2.059	2.218	2.246	100%

The centroid is comfortably below the nominal 1 ms estimator budget in this Python benchmark. The Gaussian fit is more accurate in many clean and stressed regimes, but its Python implementation does not meet a 1 kHz budget on this machine. Warm-starting reduced the median stressed latency slightly but did not meet the budget. The tail is sensitive to machine load, so these measurements are not treated as a deterministic real-time guarantee.

These measurements exclude camera transfer, image acquisition, preprocessing, synchronization, filtering, and gimbal output. They are a development baseline, not evidence of deterministic flight-hardware timing. A deployed implementation would likely use compiled C/C++ or dedicated FPGA logic, with the centroid as the natural low-latency baseline and the Gaussian fit reserved for slower supervisory, calibration, or high-confidence modes.

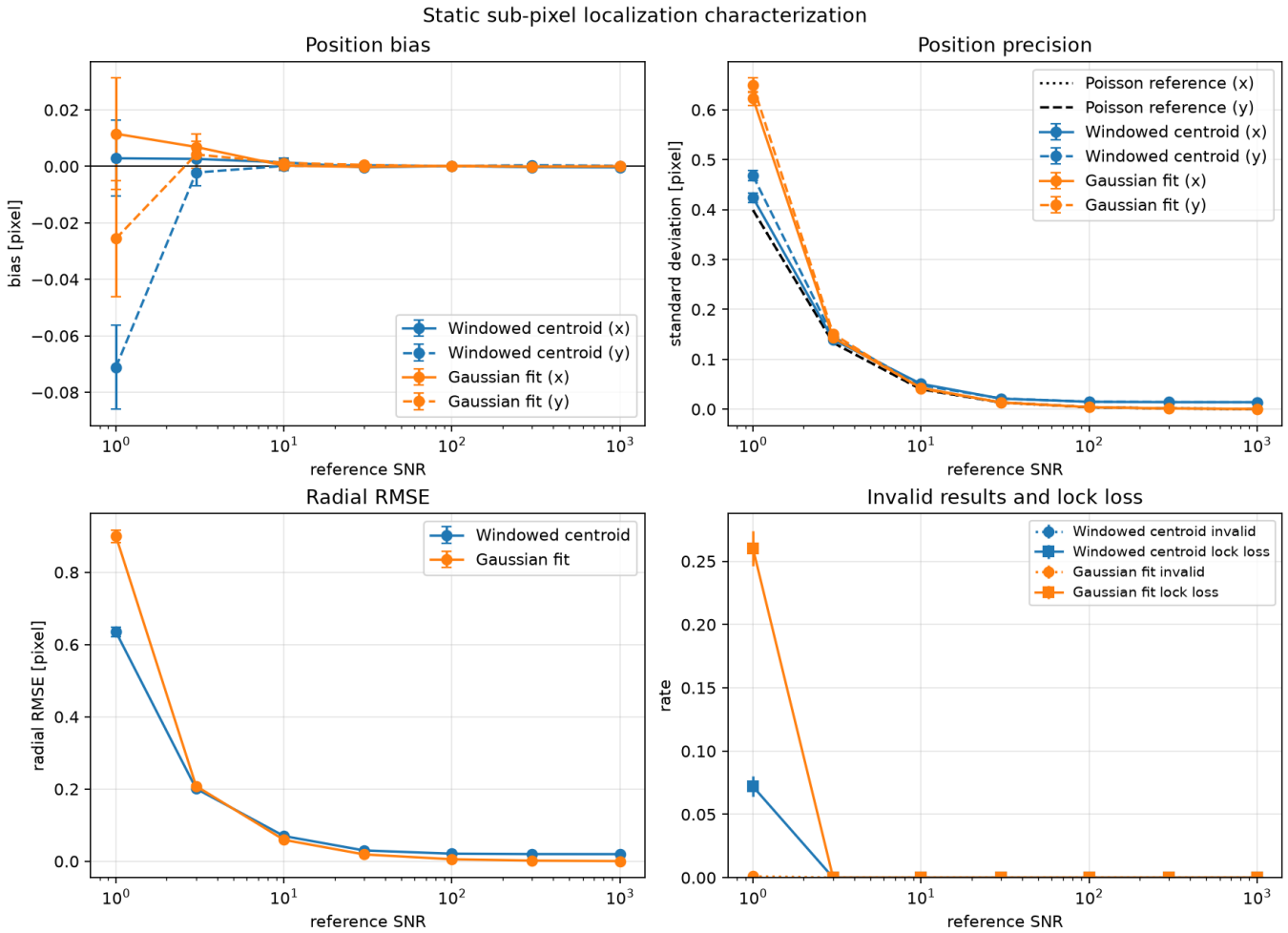


Figure 3: Ideal-model characterization and CRLB reference

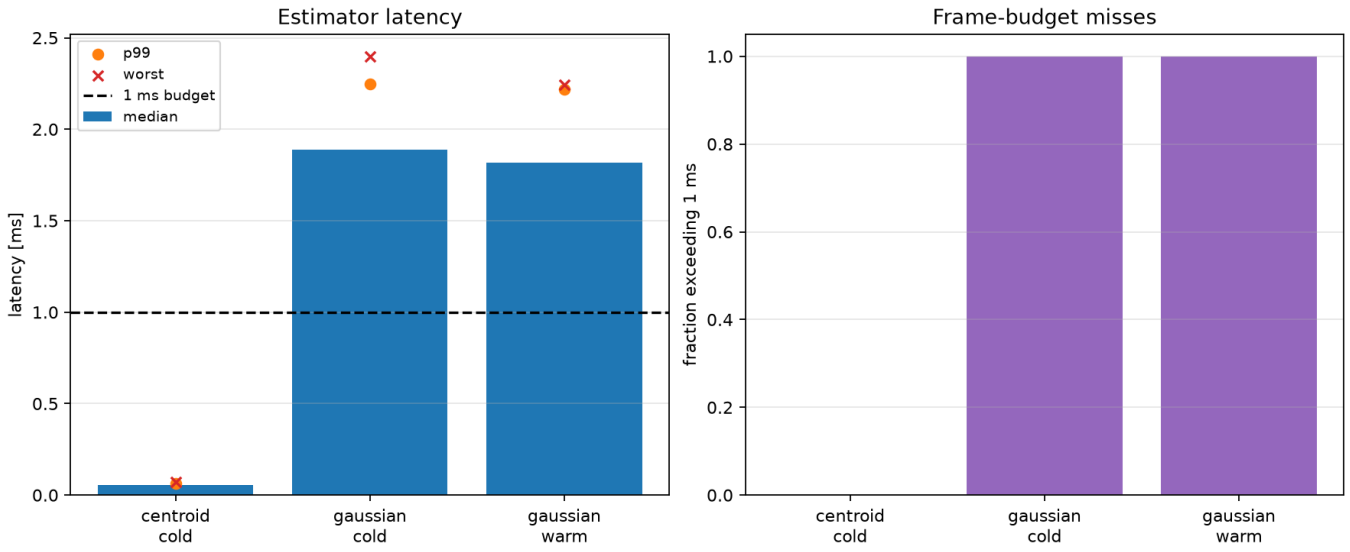


Figure 4: Estimator runtime benchmark

Figure 4. Sensor-profile estimator latency and 1 ms deadline misses. The benchmark is an estimator-only Python measurement on the development machine.

7.1 Sequential tracking-mode speed study

The primary runtime table above is an acquisition-style benchmark: frames are generated before timing at one fixed sub-pixel position, with independent sensor noise. It therefore measures estimator cost without conflating it with trajectory generation, but it is not a complete test of warm-start behavior on a moving target.

A supplementary 200-frame benchmark uses consecutive frames from the dynamic trajectory. The baseline retains the 15×15 ROI and two IRLS passes. The fast configuration uses a 13×13 ROI, one IRLS pass, a 30-evaluation optimizer limit, previous-position ROI placement, and skips centroid initialization when the previous fit is valid.

Sequential configuration	Estimator	Initialization	Median [ms]	p95 [ms]	p99 [ms]	Worst [ms]
Baseline	Gaussian	cold	1.946	2.143	2.300	2.335
Baseline	Gaussian	warm	1.814	2.073	2.152	2.266
Fast tracking	Gaussian	cold	1.035	1.201	1.312	1.356
Fast tracking	Gaussian	warm	0.972	1.147	1.260	1.300

The reduced configuration cuts median Gaussian latency by approximately 46% and makes the warm median slightly less than 1 ms, but its tail still exceeds the 1 ms budget. It is therefore a useful tracking-mode experiment, not evidence that the Python fit is deterministic at 1 kHz. A production system would still reserve the centroid for the hard real-time path or port the fit to compiled/vectorized embedded code.

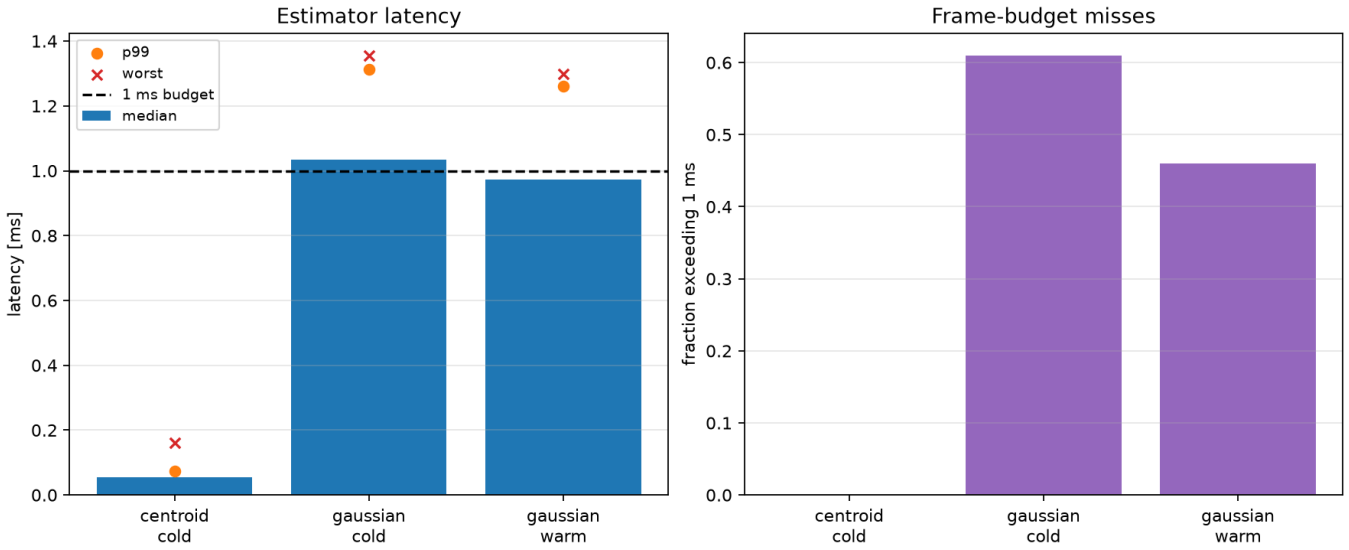


Figure 5: Sequential fast-Gaussian tracking benchmark

Figure 5. Sequential dynamic-frame benchmark for the configurable fast Gaussian tracking mode.

The raw sequential benchmark files are retained in `results/performance/gaussian_tracking_baseline/` and `results/performance/gaussian_tracking_fast/`.

7.2 Jetson Orin target validation

The final workflow was also run on an NVIDIA Jetson Orin as a bounded embedded target-validation check. The host used JetPack R36.4.3 on Ubuntu 22.04.5 (aarch64), Python 3.13.13, NumPy 2.5.3, and SciPy 1.18.1. This was the same 200-frame, 20-warm-up estimator benchmark used for the Mac reference, with the full-quality stressed profile. The implementation is CPU-only; CUDA and TensorRT were visible on the device but were not used by the code.

Estimator	Mac median [ms]	Jetson median [ms]	Jetson p99 [ms]	Jetson >1 ms
Centroid, cold	0.055	0.485	0.544	0%
Gaussian, warm	1.819	10.482	13.459	100%

The seeded static and dynamic metrics matched the Mac reference to floating- point precision, including the SNR-10 RMSE values and 73 Hz disturbance recovery. The target run therefore validates portability of the scientific results, while showing that the Python Gaussian implementation has less timing margin on the Jetson. The result remains an estimator-only measurement and does not establish end-to-end camera or gimbal latency. The saved CSV, JSON, dynamic summary, and run manifest are in `results/performance/jetson_orin/`.

8. Dynamic tracking and disturbance recovery

The final dynamic sequence contains 5,000 frames sampled at 1 kHz, giving a 5-second record and 0.2 Hz FFT-bin spacing. The target reference SNR is 10. The trajectory uses a discrete Ornstein–Uhlenbeck drift with nominal stationary RMS 0.6 pixels before the explicit bound, correlation time 0.25 seconds, and ±2-pixel drift bounds. The independent jitter has 0.04-pixel RMS and correlation coefficient 0.95. One 73 Hz sinusoidal disturbance with 0.05-pixel amplitude is injected on the x axis. The same persistent sensor maps and rendered frames are passed to both estimators.

The disturbance amplitude is 1.25 times the per-frame jitter RMS, so it is deliberately detectable but not overwhelmingly larger than the instantaneous measurement variation. Its recovered spectral peak is 14.6 times the estimated noise floor for the centroid and 25.4 times the noise floor for the Gaussian fit.

The recovered signals are not filtered before the primary result. Invalid samples would be linearly filled only for the evenly sampled spectral analysis; all trajectory-error metrics retain the original validity mask. A Hann window is applied before the FFT. Because the OU process intentionally occupies the low-frequency band, disturbance identification searches above 20 Hz, then refines the FFT peak by sinusoidal least squares including a constant and linear trend. This separates slow drift from the intentional periodic component without claiming more frequency resolution than the 5,000-sample record provides.

Estimator	RMS error [px]	95th percentile [px]	Estimated frequency [Hz]	Frequency error [Hz]	Estimated amplitude [px]	Amplitude error [px]	Peak/ noise
Windowed centroid	0.1636	0.2782	72.9934	0.0066	0.0563	0.0063	14.6
Gaussian fit	0.0817	0.1405	72.9974	0.0026	0.0565	0.0065	25.4

Both estimators produced 5,000 valid estimates with zero formal failures and zero lock loss at SNR 10. The Gaussian fit has approximately half the RMS trajectory error and a stronger recovered spectral peak, at the cost of the runtime measured in Section 7. The recovered amplitudes are slightly above the injected value, which is expected for a finite noisy record and is reported rather than corrected away. The result is shown in `dynamic_tracking.png`, and the per-frame trajectory plus summary files are saved under `results/latest/results/dynamic/`.

The primary five-second record contains one realization, so the apparent positive amplitude errors (0.0063 and 0.0065 pixel) should not be read as a calibrated systematic bias. A separate five-seed repeat used shorter 2,500-frame records to quantify repeatability. The mean recovered Gaussian amplitude was 0.05265 ± 0.00649 pixel (sample standard deviation), giving a mean bias of $+0.00265 \pm 0.00290$ pixel (standard error of the mean). This is consistent with a small positive magnitude-estimation effect within the finite-record variation, but does not establish a deterministic 13% bias. The centroid repeat had a mean amplitude of 0.04899 ± 0.00953 pixel and was less reliable for frequency identification in the shorter records. The primary 5,000-frame result remains the frequency-recovery reference.

The repeat outputs are saved under `results/latest/results/dynamic_repeats/` and the amplitude summary is shown below.

Figure 6. Five-seed amplitude repeat. Error bars show between-seed sample standard deviation; the dashed line is the injected amplitude.

The dynamic figure contains the true and recovered trajectories, radial tracking error, and the Hann-windowed disturbance spectrum. It also annotates the injected and recovered disturbance frequency and amplitude, while the CSV and JSON outputs retain the per-frame estimates and the numerical recovery metrics used in the table above.

Figure 7. True and recovered motion, radial tracking error, and recovered periodic disturbance.

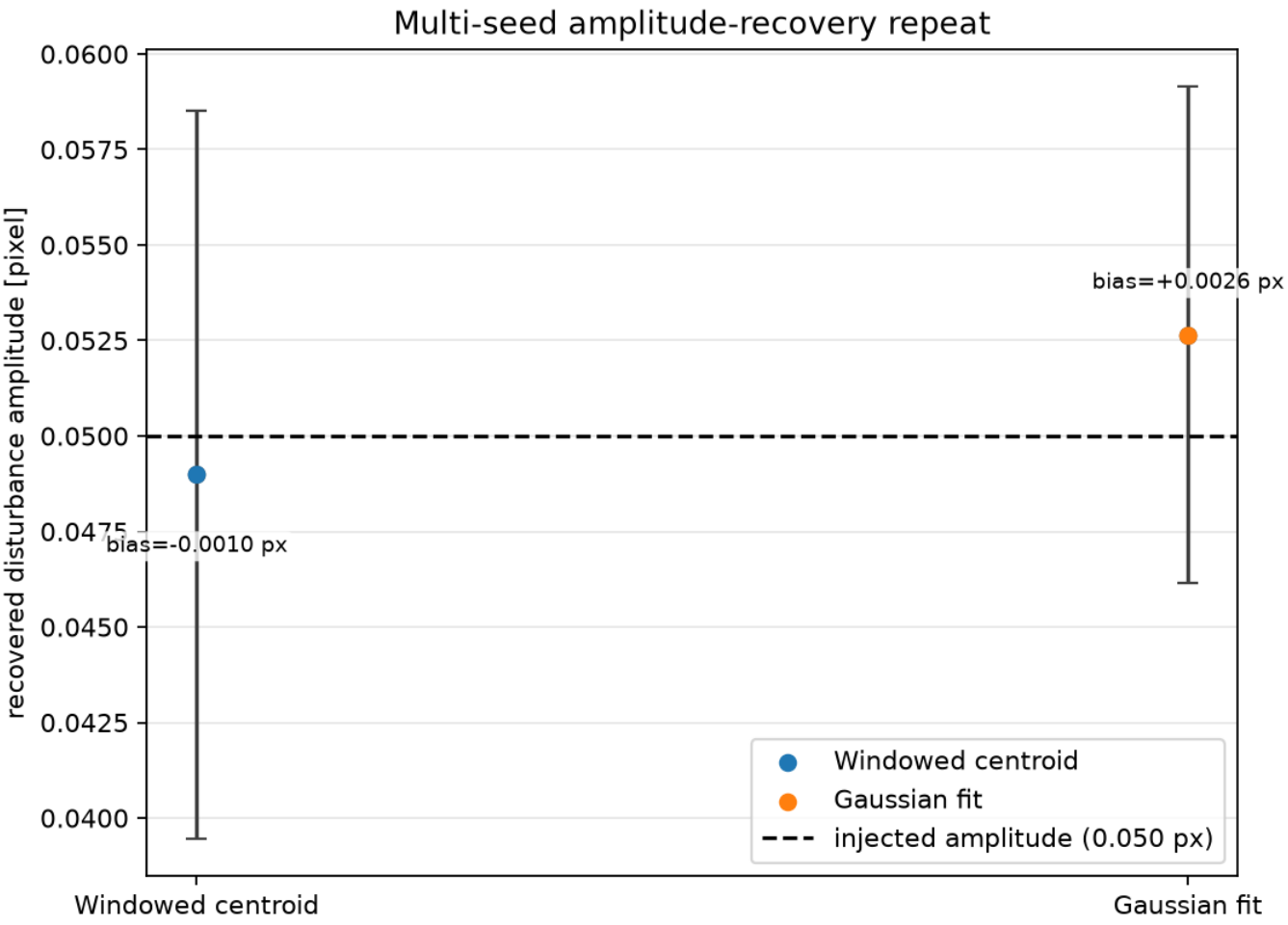


Figure 6: Multi-seed dynamic amplitude repeat

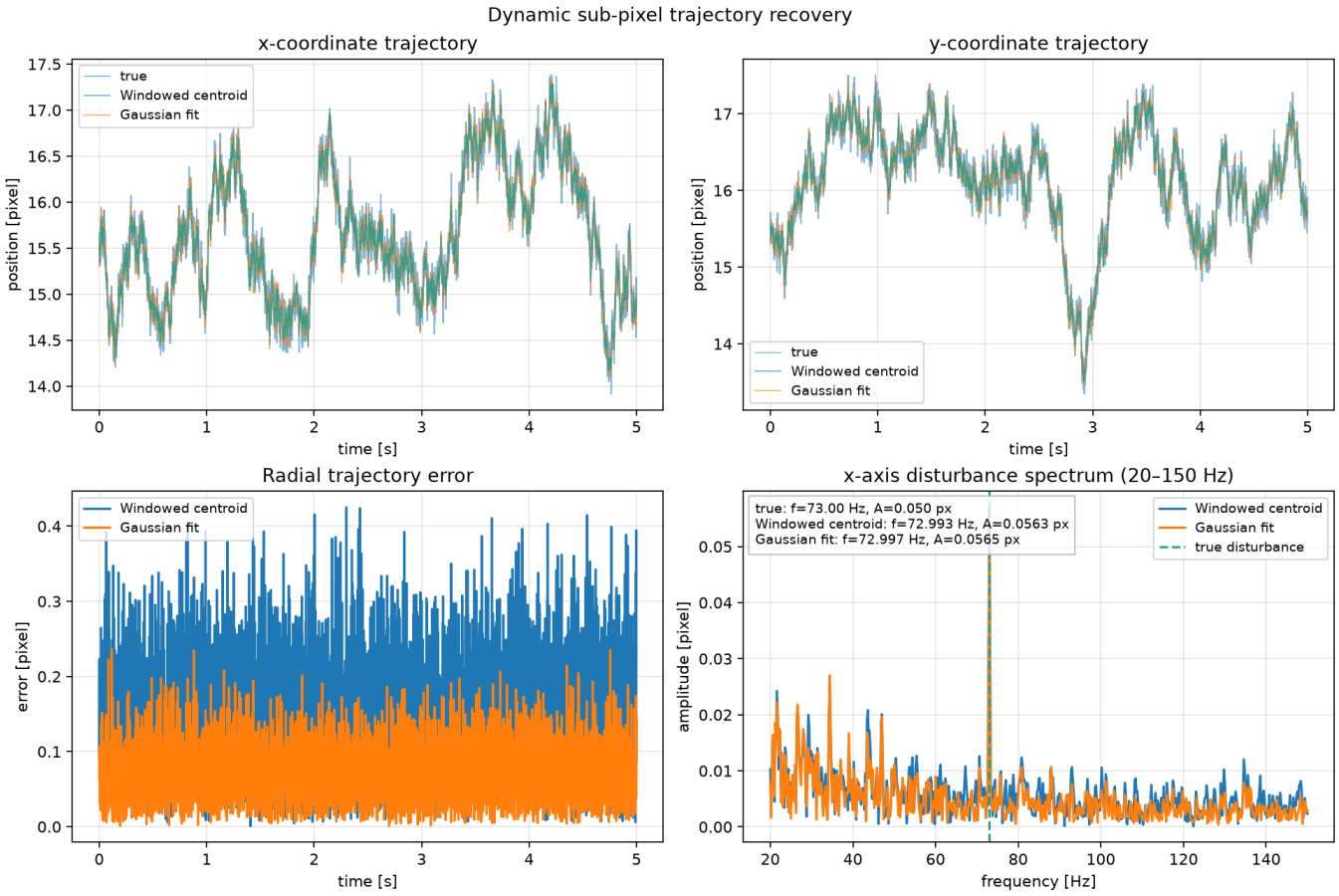


Figure 7: Dynamic trajectory and disturbance recovery

9. Outdoor conditions and robustness

The uncontrolled outdoor case is treated as a problem of separating image quality, optical motion, and timing faults. The same image signature can have different causes: a dimmer spot may indicate scintillation, attenuation, defocus, or exposure/gain error. The estimator therefore needs quality diagnostics in addition to a position estimate.

Condition	Image signature	Dominant effect on estimate	Detection signal	Handling strategy
Scintillation	Frame-to-frame spot flux changes with little corresponding centroid motion	Increased variance, intermittent low-SNR outliers, and temporal amplitude modulation	Effective peak-to-background ratio, fitted flux, short-window flux variance, estimator disagreement	Use exposure/gain control where available; gate or down-weight low-quality frames; use a causal tracker only with explicitly reported latency
Beam wander / angle-of-arrival variation	The whole PSF translates while its width and shape remain approximately stable	Real position motion, not an estimator failure; can become an outlier if it exceeds the ROI or control bandwidth	Position history, fitted width, residual quality, and consistency with platform sensors	Preserve it as a measurement; use a control-oriented tracker and ensure the ROI follows the predicted position
Thermal focus or magnification drift	PSF width and possibly total captured flux change slowly	Increased variance or model-mismatch bias; centroid and Gaussian-fit performance diverge	Fitted sigma, residual structure, and long-term flux/width trend	Track calibration versus temperature; constrain the allowed sigma range; reacquire if the fit becomes implausible
Platform vibration and exposure-time blur	Elongated or broadened PSF, often aligned with a vibration axis	Bias, increased variance, and attenuation of high-frequency motion	Second-moment/aspect-ratio residuals, fitted width, vibration spectrum, and exposure metadata	Shorten exposure when photon budget permits; model a motion-blurred PSF; do not interpret blur as a static width change
Changing sky/background illumination	Elevated background, gradients, glints, or nearby bright sources	Increased variance, background bias, false peaks, and occasional lock loss	Border background fit, peak-to-background ratio, residual plane error, and multi-peak checks	Estimate background from uncontaminated regions; mask transient bright regions; reject ambiguous multi-peak frames
Saturation, blooming, clipping, and reflections	Flat-topped spot, clipped pixels, blooming trails, or transient bright patches	Centroid bias, fit sensitivity, or dropout	Saturation fraction, clipped-pixel geometry, fitted flux, and centroid/Gaussian disagreement	Mask clipped pixels for the Gaussian fit; lower exposure/gain or use optical attenuation; declare loss of lock when usable pixels are insufficient
Dust, contamination, defocus, misalignment, or partial occlusion	Asymmetric PSF, missing tails, halos, or stable spatial artifacts	Bias and increased variance; possible persistent outliers	Residual-map asymmetry, fitted sigma, shape moments, and calibration-frame comparison	Use flat-field/bias calibration and periodic health checks; use robust fitting or reacquire when the PSF model is not credible
PRNU, dark-current drift, and hot pixels	Fixed spatial structure and isolated persistent bright pixels	Bias if the spot overlaps a bad pixel; increased variance and false peaks	Persistent bad-pixel map, dark-frame statistics, and frame-to-frame pixel consistency	Maintain dark/flat-field maps; mask known bad pixels; refresh calibration with temperature and age

Condition	Image signature	Dominant effect on estimate	Detection signal	Handling strategy
Lens distortion and calibration drift	Position-dependent geometric residuals, often smooth across the field	Spatially varying bias	Calibration-grid residuals and repeatability at known points	Apply a calibrated distortion map and monitor residuals over temperature and time
Rolling shutter or timestamp error	Position depends on row time; periodic motion appears phase-shifted or distorted	Bias and frequency/amplitude error	Exposure/readout timing metadata and row-dependent residuals	Prefer global shutter for the control path; otherwise model row timing and synchronize timestamps
Camera, estimator, filter, and actuator latency	A correct-looking estimate is temporally delayed relative to platform motion	Phase lag, reduced control margin, and possible instability	Timestamped end-to-end latency, dropped-frame counters, and closed-loop frequency response	Budget exposure, readout, transfer, estimation, filtering, and actuation separately; compensate only with measured latency

Controlled stress experiment

The repository includes a controlled experiment in `experiments/real_world_conditions.py`. It quantifies effects that can be introduced without replacing the production simulator. Each condition used 300 paired trials at a nominal reference SNR of 10, the same sub-pixel positions for both estimators, and a 1-pixel radial lock-loss threshold. For background and defocus cases, nominal total photon flux was held fixed rather than recalibrating brightness to restore SNR 10, so the loss of information is measurable.

Scintillation was represented by a stationary, mean-one correlated log-normal flux multiplier with log-intensity RMS 0.5 and lag-one correlation 0.95. This is an engineering stress model, not a claim about a particular atmospheric channel. The defocus case increased sigma from 1.75 to 2.4 pixels. The bright background case used a 200-electron/pixel offset with gradients of 1.5 and -1.0 electrons/pixel. The saturation case multiplied nominal spot flux by 100 while retaining the 12-bit ADC.

Condition	Estimator	Effective SNR (mean; 5th–95th percentile)	Lock loss	Radial RMSE [pixel]	Conditional RMSE [pixel]	Saturated-frame rate
Nominal	Windowed centroid	9.90; 8.99–10.76	0.0%	0.152	0.152	8.3%
Nominal	Gaussian fit	9.90; 8.99–10.76	0.0%	0.081	0.081	8.3%
Scintillation	Windowed centroid	8.68; 4.25–14.49	1.0%	0.268	0.247	8.0%
Scintillation	Gaussian fit	8.68; 4.25–14.49	0.0%	0.132	0.132	8.0%
Bright background	Windowed centroid	6.69; 5.87–7.50	0.0%	0.350	0.350	0.0%
Bright background	Gaussian fit	6.69; 5.87–7.50	0.0%	0.148	0.148	0.0%
Defocus	Windowed centroid	6.74; 6.64–6.85	0.0%	0.352	0.352	6.7%

Condition	Estimator	Effective SNR (mean; 5th–95th percentile)	Lock loss	Radial RMSE [pixel]	Conditional RMSE [pixel]	Saturated-frame rate
Defocus	Gaussian fit	6.74; 6.64–6.85	0.0%	0.137	0.137	6.7%
Saturation	Windowed centroid	110.06; 101.28–118.33	0.0%	0.363	0.363	100.0%
Saturation	Gaussian fit	110.06; 101.28–118.33	0.0%	0.007	0.007	100.0%

The results support three practical conclusions. First, a high nominal SNR does not guarantee good localization when the PSF is broadened or the background rises. Second, scintillation is primarily an information-quality problem: it changes effective SNR and can produce intermittent lock loss without physically moving the spot. Third, clipping should be treated as a quality condition rather than ordinary noise. The Gaussian fit was much less affected in this experiment because it excludes ADC-limit pixels, whereas the centroid remained biased by the clipped shape even when its error stayed below the formal 1-pixel lock threshold.

The simulator's saturation diagnostic marks both high-side clipping and negative pre-ADC read-noise excursions, so nominal and defocus `saturated-frame` rates are not equivalent to positive spot saturation. In the 100-times-brighter case the clipped pixels are dominated by high-side spot saturation. This distinction is retained in the saved metadata and is a known limitation of the simplified ADC diagnostic.

The controlled results are saved in `real_world_conditions.csv` and `real_world_conditions.json`, with the comparison figure in `real_world_conditions.png`.

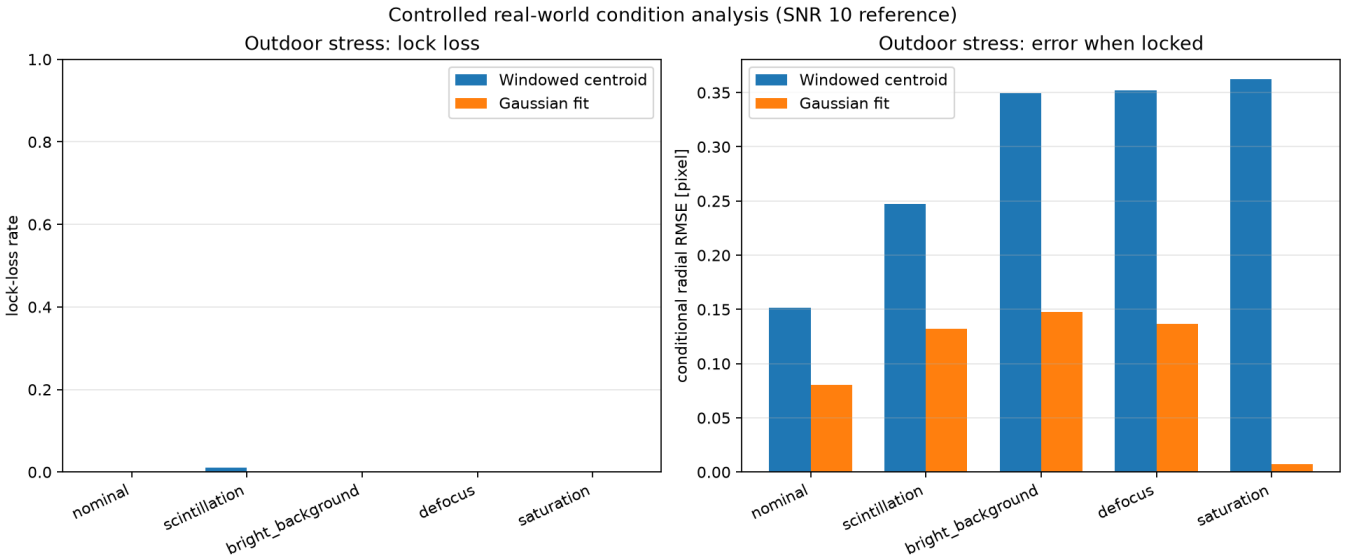


Figure 8: Controlled outdoor-condition stress comparison

Figure 8. Quantified estimator behavior under selected outdoor-condition stress cases.

10. Assumptions and limitations

The following register records the assumptions introduced so far. Each entry is also recorded in docs/decisions.md. Values are assumptions or calibrations for this exercise, not measurements of a deployed camera.

Identifier	Assumption or value	Units	Rationale	Expected effect	Validation or limitation
A-001	Pixel centers are integer coordinates and boundaries are half a pixel away	pixels	Maintains a consistent forward model	Affects phase and boundary behavior	PSF tests
A-002	Arrays use `image[y, x]`; positions use `(x, y)`	—	Separates array and Cartesian conventions	Prevents axis swaps	Foundation tests

Identifier	Assumption or value	Units	Rationale	Expected effect	Validation or limitation
A-003	The PSF is a pixel-integrated circular Gaussian	—	Represents finite pixel area	Affects phase response and theory	Flux and symmetry tests
A-004	Brightness is controlled by total expected spot electrons	electrons	Gives SNR and shot noise physical meaning	Determines photon statistics	SNR calibration
A-005	A 7-pixel $1/e^2$ diameter corresponds to sigma = 1.75 pixels	pixels	Converts the assignment convention	Sets spot width and precision	Algebra and simulator tests
A-006	Named RNG streams derive from one master seed	—	Preserves reproducibility	Controls repeatability	Same-seed tests
A-007	The nominal sensor is 32 × 32 pixels, 1 ms exposure, and 1 kHz	pixels, s, Hz	Simplified real-time scenario	Sets geometry and timing budget	Configuration validation
A-008	The primary disturbance is 73 Hz	Hz	Aligns with 2,000- and 5,000-frame FFT bins at 1 kHz	Simplifies the baseline spectral result	Sampling-grid calculation
A-009	PRNU multiplies expected total charge before Poisson sampling	gain	Explicit sensor-response convention	Changes means and shot-noise variance	Ablation tests; model limitation
A-010	Hot pixels have persistent elevated dark-current rates	electrons/s	Captures fixed spatial outliers	Adds persistent stochastic contamination	Fixed-map tests
A-011	Read noise is additive Gaussian noise in electrons; quantization rounds to nearest DN	electrons RMS, DN	Standard simplified readout model	Controls low-light behavior	Noise and ADC tests
A-012	Reference SNR uses a pixel-centered spot; effective SNR is also recorded	dimensionless	Avoids phase-dependent SNR labels	Separates sweep control from actual frame information	SNR tests and metadata
A-013	Stressed sigma variation is 5% RMS about 1.75 pixels	pixels	Represents modest optical variation	Produces model mismatch	Frame metadata; later sensitivity study
A-014	Default stressed parameters are 1% PRNU, 50 electrons/s dark current, four 5,000-electron/s hot pixels, 3-electron read noise, 12-bit ADC, and 1 electron/DN	mixed	Controlled engineering assumptions	Sets stress and saturation regime	Final sweep, ROI study, and controlled outdoor analysis
A-015	Major noise effects have independent switches	—	Enables ablation studies	Supports causal diagnosis	Toggle tests
A-016	The visual sanity tool reuses the production simulator	—	Prevents a second visualization model	Improves debugging confidence	Headless figure generation
A-017	Both estimators use full-frame peak detection and local odd-sized ROIs; defaults are 13 × 13 and 15 × 15	pixels	Provides a fair input contract	Affects bias, variance, and runtime	Estimator tests; stressed ROI sensitivity study
A-018	Centroid background is fitted on the ROI border; negative residuals are clipped	DN, pixels	Handles the nominal background gradient	Can remove weak PSF tails	Clean test and ROI-sensitivity study
A-019	The Gaussian fit estimates position, flux, width, and planar background with the pixel-integrated model	DN, pixels	Matches the simulator and estimates nuisance terms	Model mismatch affects bias and convergence	Jacobian and clean-fit tests
A-020	Gaussian fitting uses bounded least squares, two IRLS steps, ADC-limit masking, and a configurable 1 DN^2 variance floor	mixed	Controls optimizer behavior and prevents extreme tail-pixel weights	Approximate weighting affects stressed results	Jacobian and saturation tests
A-021	The GUI reuses the production simulator, trajectory generator, and estimators	—	Keeps the demonstration consistent	GUI is not authoritative scientific output	Headless GUI test
A-022	GUI display timing defaults to 50 ms and is not a 1 kHz benchmark	ms, Hz	Makes the display usable	Separates visualization from latency evidence	Dedicated runtime benchmark complete
A-023	The GUI sanity mode uses 1.5-pixel, 0.4-Hz sinusoidal drift, 0.04-pixel correlated jitter, and a 0.05-pixel 73-Hz disturbance	pixels, Hz	Provides an easily inspected deterministic motion mode	It is not the final stochastic dynamic model	Trajectory reproducibility test
A-024	The final dynamic drift is a discrete OU/AR(1) process with nominal 0.6-pixel stationary RMS before bounding, 0.25-second correlation time, ±2-pixel drift bounds, and the saved master seed	pixels, s	Avoids unbounded Brownian drift and isolates one intentional periodic disturbance	Drift statistics affect recovery and spectral analysis	5,000-frame dynamic sequence
A-025	Static SNR sweeps reuse trial positions and fixed-pattern maps across SNR values, and both estimators process the same rendered frames	—	Reduces avoidable Monte Carlo comparison variance	Makes estimator comparisons more paired and reproducible	Final ideal and stressed sweeps
A-026	Gaussian IRLS uses a configurable 1 DN^2 lower bound on approximate Poisson/read-noise variance	DN^2	Prevents extreme weights from tiny or negative fitted tail-background values	Changes approximate fit weighting in low-signal tails	Final sweeps; likelihood remains approximate
A-027	Runtime frames are generated before timing; 20 warm-up calls are excluded; centroid cold and Gaussian cold/warm paths are timed against a 1 ms interval	ms, frames	Separates estimator execution from simulation and initialization overhead	Determines reported latency and deadline-miss rates	200-frame ideal and stressed benchmark
A-028	The final dynamic record uses 5,000 frames at 1 kHz, 0.04-pixel correlated jitter, a 73 Hz/0.05-pixel x-axis disturbance, Hann-windowed spectral analysis, and a 20 Hz lower search limit	frames, Hz, pixels	Provides frequency resolution while separating slow OU drift from the intentional disturbance	Analysis results depend on windowing, search band, and finite-record noise	Dynamic recovery output and 44-test suite
A-029	Phase 6 controlled stress cases use 300 paired trials at reference SNR 10, a 1-pixel lock-loss threshold, and fixed nominal photon flux for background/defocus comparisons	frames, dimensionless, pixels	Makes outdoor-condition effects measurable without silently restoring baseline SNR	Results are finite-sample engineering evidence, not field validation	Real-world CSV/JSON outputs and reproducibility test

Identifier	Assumption or value	Units	Rationale	Expected effect	Validation or limitation
A-030	Scintillation is a mean-one correlated log-normal flux multiplier with log-intensity RMS 0.5 and lag-one correlation 0.95	dimensionless	Captures temporal intensity modulation while preserving the distinction between flux variation and beam wander	Changes effective SNR and can create intermittent lock loss	Controlled Phase 6 experiment; atmospheric statistics are not measured
A-031	The controlled outdoor study uses PSF sigma 2.4 pixels for defocus, a 200-electron/pixel bright background with 1.5/-1.0 electron/pixel gradients, and 100x nominal flux for saturation	pixels, electrons, electrons/pixel, multiplier	Provides interpretable stress levels within the existing sensor model	Quantifies sensitivity to width, background, and clipping; does not model blooming or optical halos	Saved condition metadata and figure
A-032	GUI keyboard demonstrations are display-only; manual nudges are 0.1 pixels, flux presets use SNR 1 and 1,000, and the forced hot pixel is fixed at a Gaussian-ROI edge	pixels, dimensionless	Makes phase, saturation, gradient, and outlier behavior easy to demonstrate during review without changing reproducible experiments	Interactive behavior is illustrative and depends on figure focus	GUI shortcut test; offline results remain authoritative
A-033	Fast Gaussian tracking uses a 13 × 13 ROI, one IRLS pass, a 30-evaluation optimizer limit, previous-position ROI placement, and previous-position initialization	pixels, passes, evaluations	Provides a configurable speed study without changing the full-quality acquisition defaults	Reduces latency but can increase model truncation, convergence, or tracking failure risk	Sequential runtime benchmark; not the production default
A-034	Acquisition and sequential tracking runtime benchmarks are reported separately	—	Prevents fixed-position estimator timing from being mistaken for deployed warm-start behavior	Makes warm-start conclusions conditional on input sequence	200-frame fixed and dynamic benchmark artifacts
A-035	The ranking ablation uses 250 paired trials at SNR 1 and 3 for five conditions: ideal, background gradient, PRNU/hot pixels, quantization/saturation, and full stressed	trials, dimensionless	Separates low-SNR convergence effects from contamination and clipping effects	Provides evidence for the clean/stressed ranking reversal	Ablation outputs and Figure 2
A-036	Dynamic amplitude repeat uses five independent master seeds and 2,500 frames per seed at target SNR 10	seeds, frames, dimensionless	Quantifies finite-record amplitude variation without replacing the primary 5,000-frame result	Gives uncertainty on the apparent amplitude bias; short records can weaken frequency recovery	Repeat outputs and Figure 6
A-037	Sensor-chain validation includes a non-unity 2-electron/DN conversion-gain case	electrons/DN	Ensures the nominal 1-electron/DN choice does not mask unit-ordering errors	Separates charge-domain clipping from DN conversion	ADC conversion-gain test
A-038	GUI-only scene sweep uses independent beacon/background scaling with bounded exposure and simplified analog-gain response	dimensionless, ms, gain	Demonstrates the optional day/night exposure concept without changing authoritative experiments	Shows saturation recovery and low-light response qualitatively	Headless GUI test; interactive review only
A-039	Jetson Orin validation uses the CPU NumPy/SciPy implementation and the same 200-frame, 20-warm-up estimator benchmark as the Mac reference	frames, ms	Separates embedded-target portability evidence from the Mac reference while keeping the input contract paired	Measures platform-dependent estimator timing; does not cover camera, transfer, or gimbal latency	Jetson runtime/dynamic artifacts; machine-load dependent

11. Reproducibility and implementation status

11.1 Implementation details and reproducibility record

The final figures and numerical outputs were regenerated under the `.venv` environment on the following development system:

Item	Recorded value
Host	MacBook Pro, model Mac15,6
Processor	Apple M3 Pro, 11 cores (5 performance, 6 efficiency)
Memory	18 GB
Operating system	macOS 27.0, arm64
Python environment	`.venv`, Python 3.13.1
Core packages	NumPy 2.5.3; SciPy 1.18.1; Matplotlib 3.11.2; PyYAML 6.0.3; pytest 9.1.1
Final run	`20260927T080135Z`
Experiment sizes	1,000 static trials/SNR; 200 ROI trials; 200 runtime frames plus 20 warm-up calls; 5,000 dynamic frames; 300 outdoor-condition trials; 250 ablation trials/condition/SNR; five 2,500-frame amplitude repeats
Reproduction command	`python run_all.py --config configs/final.yaml`

The run manifest records the configuration, software versions, seed, and generated files in the following file:

`results/latest/run_manifest.json`

Seeded simulation outputs are deterministic; runtime measurements remain dependent on machine load and operating-system scheduling.

The Mac record above is the authoritative reference run for the checked-in figures and seeded results. An additional target-validation run was completed on an NVIDIA Jetson Orin (JetPack R36.4.3, Ubuntu 22.04.5, aarch64, Python 3.13.13, NumPy 2.5.3, SciPy 1.18.1). Its runtime and dynamic summary artifacts are preserved in `results/performance/jetson_orin/`. The two hosts produced the same seeded accuracy and dynamic metrics; only the machine-dependent estimator timing changed materially.

The repository uses a standard Python virtual environment and requires NumPy, SciPy, Matplotlib, PyYAML, and pytest. The setup commands are documented in the README. After activating `.venv`, the test command is:

```
python -m pytest -q
```

The controlled real-world conditions analysis can be reproduced with:

```
python -m experiments.real_world_conditions \
  --trials 300 \
  --output-dir results/real_world \
  --figure results/standalone/figures/real_world_conditions.png
```

The live demonstration can be launched with:

```
python -m experiments.live_gui --profile sensor --snr 10
```

The GUI is an optional review aid. It is not a replacement for the required Monte Carlo plots, dynamic results, runtime measurements, or one-command reproduction workflow. The complete workflow is:

```
python run_all.py --config configs/final.yaml
```

It regenerates the final static, ROI, runtime, dynamic, ablation, repeat, and real-world outputs under a UTC timestamped directory in `results/runs/`, then refreshes `results/latest/` as a stable snapshot and writes its manifest. Runtime measurements are machine-load dependent; the seeded simulation products are deterministic. Optional YAML files such as `configs/final.yaml` provide validated overrides for pipeline controls and individual scenario profiles; omitted values continue to inherit from `config.py`.

For live review, the figure also supports keyboard-driven demonstrations: Arrow keys apply 0.1-pixel target nudges, Shift+S and Shift+D select low- and high-flux presets, G toggles the configured background gradient, H adds one persistent hot pixel at the edge of the Gaussian ROI, and Q/Escape closes the window. Matplotlib's default keyboard shortcuts are disabled for this figure. The hot-pixel case is a controlled GUI intervention rather than a change to the production sensor model, and is used to discuss estimator sensitivity rather than claim automatic rejection by the Gaussian fit.

The GUI also includes a separate scene-brightness control spanning a nominal night-to-day illumination sweep. Beacon flux and ambient background are varied independently; the midpoint is the nominal sensor scenario. An optional automatic exposure/gain mode responds to saturated or underexposed frames with bounded, hysteretic updates and reports the current exposure, gain, effective SNR, saturation count, and lock status. This is a qualitative review feature, not a calibrated camera-control implementation, and it does not modify the authoritative Monte Carlo, runtime, or dynamic results.

Figure 9. Live simulator demonstration using the production simulator and estimators, including the optional illumination sweep and exposure/gain response. The Markdown report displays the animation; PDF export uses its first-frame PNG because PDF does not provide animation. The display is intentionally slower than the 1 kHz benchmark and is included as a review aid rather than as a quantitative result.

11.2 Reproduction pipeline

With the final YAML configuration, `run_all.py` executes the following stages in order. The first eight stages are scientific analyses; the final stage writes provenance metadata and refreshes the stable `results/latest/` snapshot. The ablation and multi-seed repeat stages are enabled by `configs/final.yaml` but can be disabled for smaller runs.

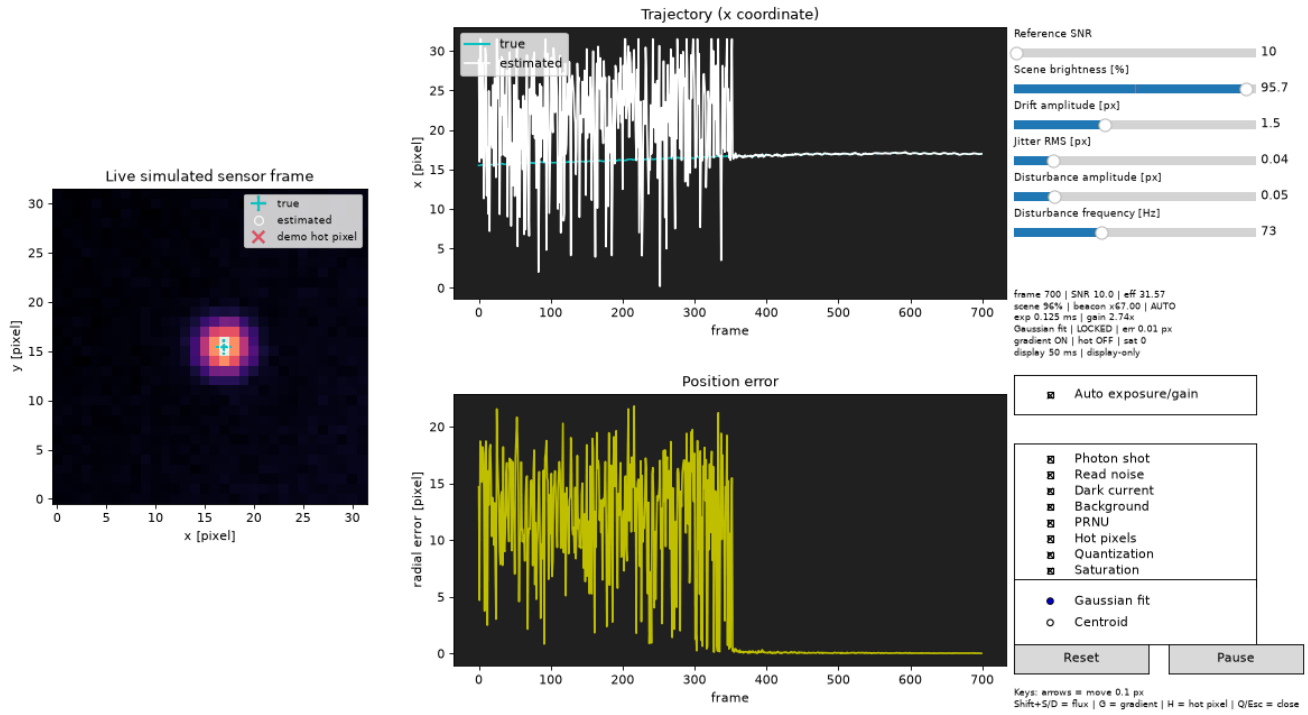


Figure 9: Animated live simulator demonstration

Stage	Operation
1	Ideal SNR characterization and Fisher reference
2	Controlled estimator ablation
3	Stressed-sensor SNR characterization
4	Multi-seed dynamic amplitude repeat
5	ROI sensitivity study
6	Estimator runtime benchmarks
7	Dynamic trajectory and disturbance recovery
8	Controlled outdoor-condition stress analysis
9	Manifest creation and latest-snapshot refresh

12. Assignment traceability

Assignment requirement	Evidence	Status
Configurable Gaussian spot at a known sub-pixel position	Sections 2.1–2.2; PSF module; tests	Implemented
Photon, read, dark, hot-pixel, gradient, PRNU, and quantization noise	Section 2.3; sensor-noise module; tests	Implemented
SNR control and SNR sweep	Section 2.2; configuration; SNR experiment	Complete; 1,000-trial results saved
At least two centroiding methods	Section 3; estimator package	Implemented and characterized
Bias and standard deviation versus SNR	Section 5; SNR experiment; characterization outputs	Complete; ideal and stressed figures saved
Theoretical precision reference	Section 6; theory experiment; characterization outputs	Complete for clean reference model
Per-frame compute cost and 1 kHz tradeoff	Section 7; runtime experiment; Mac reference and Jetson target-validation outputs	Complete estimator-only Python baseline; end-to-end hardware latency remains out of scope
Configurable Gaussian tracking speed study	Section 7.1; fast-Gaussian profile; performance outputs	Complete supplemental benchmark; fast profile remains optional
Estimator ranking ablation	Section 5.1; ablation experiment; ablation outputs	Complete; controlled causal comparison at SNR 1 and 3

Assignment requirement	Evidence	Status
Moving spot with drift, jitter, and periodic disturbance	Section 8; trajectory and dynamic experiment	Complete; 5,000-frame OU sequence saved
Recovered trajectory error	Section 8; dynamic summary and trajectory outputs	Complete
Recovered disturbance frequency and amplitude	Section 8; dynamic spectrum, summary, and multi-seed repeat outputs	Complete
Challenging parameters and failure modes	Sections 3, 5, 8, 9, and 10	Complete; controlled stress results and outdoor handling table included
Outdoor real-world conditions and handling	Section 9; outdoor-conditions experiment	Complete analysis; selected effects quantified
One-command reproducibility	Section 11; `run_all.py`; run manifest and run archive	Complete; production workflow verified
Written report	This document	Complete through final integration; runtime caveat documented
Reproducible dynamic scenario and analysis	Section 8; dynamic outputs and repeat outputs	Complete
Optional live interactive demonstration	Section 11; live GUI module	Implemented
Optional automatic exposure/gain demonstration	Section 11; live GUI scene sweep and exposure controller	Implemented as a qualitative GUI-only extension; not hardware validated

13. Conclusion and future work

This work developed and evaluated a configurable sub-pixel spot-localization pipeline using a pixel-integrated Gaussian sensor model, a windowed intensity centroid, and a pixel-integrated Gaussian fit. At SNR 10 in the stressed sensor profile, the radial RMSE was 0.161 pixels for the centroid and 0.081 pixels for the Gaussian fit. At SNR 1, lock-loss rates were 96.5% and 90.1%, respectively, demonstrating that improved conditional accuracy does not imply reliable localization once the signal becomes unobservable. The ideal-model Fisher reference at SNR 10 was 0.0564 pixels.

The dynamic 5,000-frame experiment produced RMS trajectory errors of 0.1636 pixels for the centroid and 0.0817 pixels for the Gaussian fit. The injected 73 Hz disturbance was recovered at 72.9934 Hz and 72.9974 Hz, with amplitudes of 0.0563 and 0.0565 pixels against the injected 0.05-pixel value. The centroid latency was 0.055 ms on the Mac reference system and 0.485 ms on the Jetson Orin, while the full Gaussian fit required 1.819 ms and 10.482 ms, respectively. These results support using the centroid as the low-latency tracking estimator and reserving the Gaussian fit for slower supervisory processing or a compiled embedded implementation.

The study provides a reproducible baseline through the saved experiment outputs, tests, report figures, and one-command workflow. The runtime results are estimator-only measurements; the complete photon-to-actuator latency, including exposure, readout, transfer, filtering, and gimbal actuation, was not measured. Future work should validate the noise model against camera calibration data, derive a precision reference for quantized and clipped likelihoods, model motion blur and rolling-shutter timing, and evaluate a compiled Gaussian implementation with hardware-in-the-loop measurement of the full control-path latency.